

AEGIS Architecture

Anonymization & Exchange Gateway for Imaging Studies

SENDING SOURCES External Browser Upload · DICOM Network (C-STORE SCP, port 11112) · Programmatic Batch Ingest

React Upload Portal (PWA)

- Drag-and-drop: files, folders, DICOM directories
- Client-side tag de-identification (DICOM PS3.15)
- Anonymization preview — before / after tag diff
- Multi-study detection with per-study upload cards
- Per-project retained-tag anonymization profiles
- Auto-retry: 3× exponential backoff per file

DIMSE Receiver (pynetdicom · GCE VM)

- Compute Engine VM — aegis-prod-dimse-receiver
- Static IP 35.232.172.221 · TCP port 11112
- C-STORE SCP — receives from PACS systems / scanners
- Institution attribution (AE title or IP CIDR)
- Calls POST /api/ingest on DICOM association close
- Durable retry queue + dead-letter (disk-persistent)

Key Privacy Principle

PHI is stripped in the browser BEFORE data leaves the hospital.

- Only tag-de-identified DICOM is transmitted
- HIPAA Safe Harbor — 18 identifier types removed
- Deterministic UID hashing + date shifting
- Zero software install required at sending site
- DIMSE path ingests raw files (internal-only path)

GCP PROJECT aegis-prod-488120 · us-central1 · Live, February 2026

Cloud Armor (DDoS / WAF) + Global HTTPS Load Balancer + Identity-Aware Proxy (admin routes) + Managed SSL cert v3 (api · admin · aegisimaging.ai · www) + Cloud Run (API · Dashboard · Landing · 8 sidecars)

API Backend (Go / Cloud Run)

- Upload orchestration (sessions, chunked PUT)
- Study / project / institution management
- Routing rules engine → auto-pipeline dispatch
- DICOMweb proxy (QIDO-RS + WADO-RS)
- Export shares: token-auth, ZIP download
- Email digest scheduler + SMTP relay
- DIMSE retry control proxy
- MCP tool backend + batch import CLI
- distroless image — minimal CVE surface

Admin Dashboard + Weasis DWV (React / Cloud Run)

- Behind Identity-Aware Proxy (IAP)
- Study browser: filter, search, paginate, bulk ops
- Weasis DWV — yoked before/after defacing review
- 7-stage pipeline visualization per study
- RBAC: admin (write) + viewer (read-only)
- Routing rules, institutions, anon profiles
- Protocol templates, API keys, webhooks
- Audit log + CSV export, share management
- Federation peers, project lifecycle

Cloud SQL (PostgreSQL 15)

- Studies, projects, institutions, admin users
- Routing rules, audit trail, export shares
- Private IP · Secret Manager credentials
- Point-in-time recovery (7-day retention)

Cloud Storage (GCS)

- dicom/raw/{uid}/ — tag-de-identified
- dicom/clean/{uid}/ — defaced + processed
- bids/{uid}/ — NIFTI / BIDS output
- Shared volume (Go API + all 8 sidecars)

Processing Sidecars (Python / Cloud Run) — dispatched async by Go API, runs hands-free via routing rules

Defacing (Python)

- Head MRI / PET / CT — facial feature removal
- mri_reface / DeepDefacer / mri_deface backends
- nibabel fallback for dev / testing only
- Writes defaced DICOM to clean/ store
- SSIM-based QA score per study

PHI Detection (Python)

- Burned-in text OCR on DICOM pixel data
- Tesseract (local) / Cloud Vision / Textract
- Per-file findings with confidence scores
- Per-project confidence threshold config
- Informational — non-blocking to pipeline

QC Automation (Python)

- File integrity + required DICOM tag check
- Slice consistency (rows / cols / pixel spacing)
- SNR estimation (signal mean / corner noise)
- Coverage completeness by body part
- Missing slice gap detection

Classification (Python)

- Fills modality + body_part from DICOM headers
- Heuristic: tags → SOP UID → series description
- Cloud Vision / Rekognition image fallback
- Re-evaluates routing rules after classify
- Confidence threshold gating (≥ 0.5)

BIDS Conversion (Python)

- DICOM → NIFTI via dcm2nii
- BIDS-compliant directory structure
- sub-[hash8]/anat | func | dwi | perf | pet/
- JSON sidecar metadata per series
- Privacy: UID-hashed subject labels
- ZIP download via Go API

Protocol Check (Python)

- Verifies TR / TE / flip / thickness vs template
- Classic + Enhanced DICOM (multi-frame)
- Per-project protocol templates with CRUD
- numeric / exact / range / contains_all match
- critical / warning / info severity levels

Synth MRI (Python) ★

- Synthetic brain MRI generation
- CPU: Shepp-Logan phantom (numpy + pydicom)
- GPU: MONAI BraTS LDM (INCLUDE_MONAI=true)
- T1w contrast, Rician noise, face anatomy
- Seeded + reproducible, 20 – 200 slices
- Imports via normal pipeline (auto-dispatch)

Security Layers

- VPC private subnets + Cloud NAT
- Cloud Armor DDoS / WAF
- TLS 1.2+ on all endpoints
- CMEK (Cloud KMS)
- IAM least privilege
- Secret Manager (DB creds, API keys)
- Cloud Audit Logs
- Artifact Registry vuln scanning
- distroless containers (Go API)
- IAP on all admin routes
- No PHI in email / audit entries

Export & Sharing

Export Portal (React / public SPA)

- Token-authenticated share links
- Study info, expiry countdown (server-anchored)
- ZIP download of approved DICOM files

DICOM Forwarding (route_to)

- DICOMweb STOW-RS destinations
- DIMSE C-STORE to remote AE Title
- Auto-forwards on study approval
- Cross-cloud: GCP→AWS STOW-RS live

Share Lifecycle

- Extend / revoke anytime
- Immutable download log per share
- Export analytics dashboard

MCP Server + AI Agent Tools

Model Context Protocol (Claude / Cursor)

- 52+ read tools: studies, audit, stats, routing, health
- 29+ write tools (confirm:true + reason required)
- Zod-validated schemas — read-safe by default
- DIMSE retry proxy: process, replay, dead-letter

Agent Orchestrator

- DICOM tag provenance analysis
- 9 diagnostic tools for study triage
- Cohort report, pipeline funnel, project health

Batch Import + Webhooks + API Keys

- aegis-import CLI — bulk historical migration
- 5 webhook events, HMAC-SHA256 signed payloads
- Machine-to-machine bearer tokens (hashed)

Landing Page (nginx / Cloud Run)

- aegisimaging.ai — public marketing site
- React + Vite built, served by nginx
- Interactive synthetic DICOM demo widget
- Schedule Demo / Contact form
- Market opportunity: \$45B imaging market
- LB default backend — no IAP required
- www + apex domains on SSL cert v3

AWS ACCOUNT 301691475234 · us-east-1 · ✓ Live — aws.api.aegisimaging.ai · Cross-cloud routing verified · Feb 25, 2026

ALB + Cognito (Auth Layer)

- HTTPS listener on ACM certificate
- Cognito hosted UI — admin-create-only user pool
- authenticate-cognito default action
- Public bypass rules: /healthz, upload, export

RDS + S3 (Data Layer)

- RDS PostgreSQL 15 — private subnet
- Secrets Manager — master credentials
- S3 DICOM bucket — versioned, KMS-encrypted
- Same STORAGE_MODE=s3 as GCP (same Go code)

API + Admin (ECS Fargate)

- Go API — same image as GCP (1 vCPU / 2 GB)
- Admin Dashboard — React / nginx (0.5 vCPU / 1 GB)
- Service discovery: api.aegis.local
- Force-new-deployment via GitHub Actions

DIMSE EC2 (t3.small)

- Elastic IP — stable for PACS AE title registration
- Amazon Linux 2023 — Docker + SSM agent
- SSM Parameter Store → image URI on every boot
- GitHub Actions: write SSM param + reboot instance

7 Python Sidecars (ECS Fargate)

- defacing · phi-detection · qc-service
- bids-service · classification-service
- protocol-service · synth-service
- Cloud Map private DNS: svc.aegis.local:8080

GitHub Actions CI/CD

- Triggers on push to develop — parallel with GCP Cloud Build
- Matrix build: 11 services, --platform linux/amd64
- Push SHA tag + latest tag to 13 ECR repositories
- Force-new-deployment for 10 ECS services
- Update SSM param + reboot DIMSE EC2

AZURE SUBSCRIPTION Container Apps · PostgreSQL Flexible Server · Azure Blob Storage · ✓ Live — Feb 26, 2026 · GitHub Actions OIDC CI/CD

Azure Container Registry + Easy Auth

- ACR — private registry, GitHub Actions OIDC push
- Easy Auth on Container Apps — injects X-MS-CLIENT-PRINCIPAL-NAME
- AUTH_PROVIDER=azure, AUTH_ENABLED=true
- Federated OIDC — no long-lived credentials stored

PostgreSQL Flex + Azure Blob (Data Layer)

- Azure Database for PostgreSQL — Flexible Server
- Azure Blob Storage — STORAGE_MODE=azure
- DefaultAzureCredential — Workload Identity / Managed Identity
- SAS tokens via user-delegation key for signed URLs

API + Admin (Azure Container Apps)

- Go API — same image as GCP/AWS (0.5-1 vCPU / 2 GB)
- Admin Dashboard — React / nginx Container App
- Azure Communication Services SMTP relay (smtp.azurecomm.net)
- Force-new-revision via GitHub Actions on every push

DIMSE Azure VM (Standard_B2s)

- Debian 12 — Docker + Azure VM Extensions
- Static public IP — TCP port 11112 for PACS registration
- GitHub Actions: az vm run-command + docker pull/restart
- Same pynetdicom C-STORE SCP as GCP/AWS

7 Python Sidecars (Container Apps)

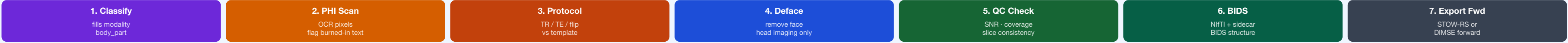
- defacing · phi-detection · qc-service
- bids-service · classification-service
- protocol-service · synth-service
- Internal ingress only — same Docker images as GCP/AWS

GitHub Actions CI/CD (OIDC)

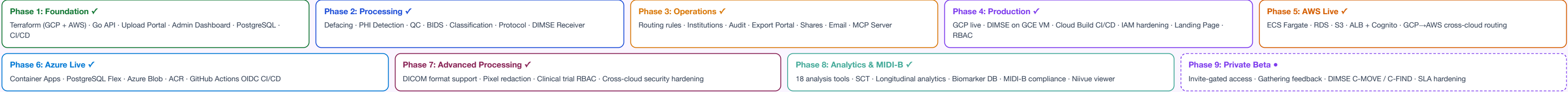
- Triggers on push to develop — parallel with GCP + AWS
- Federated OIDC — AZURE_CLIENT_ID / TENANT_ID / SUBSCRIPTION_ID
- Build 13 images → push to ACR → az containerapp update
- deploy-dimse job: az vm run-command on DIMSE VM

Automated Processing Pipeline

routing rules define which steps are required → pipeline dispatches each step in dependency order, hands-free



Implementation Phases



Key Open-Source Dependencies

Go:

Browser:

Defacing:

PHI/OCR:

QC/BIDS:

Analytics:

SCT/Spine:

Synth:

Infra:

suyashkumar/dicom · pgx · testcontainers-go · testify

dcmjs · dicomParser · Weasis DWV (MIT) · React 19 · Vite

mri_reface · DeepDefacer · mri_deface · dcm2niix · nibabel

pytesseract · Google Cloud Vision · AWS Textract · Pillow

pydicom · numpy · dcm2niix · pynetdicom (C-STORE SCP)

FreeSurfer · FSL · ANTs · SPM · SynthSeg · nnU-Net · TotalSegmentator · MONAI Label

Spinal Cord Toolbox · TotalSpineSeg · SPINEPS · MedSAM2

numpy · pydicom · MONAI Generative · torch · huggingface-hub

Terraform Google + AWS + Azure providers · Docker distroless/slim

Multi-Cloud + Test Coverage

GCP:

AWS:

Local:

Auth:

Storage:

Azure:

CI/CD:

Domains:

Cloud Run · Cloud SQL · GCS · IAP · Cloud Armor · KMS · Pub/Sub

ECS Fargate · RDS · S3 · ALB + Cognito · ACM · Secrets Manager

Docker Compose · PostgreSQL 15 · Mailpit · local filesystem

GCP IAP · Azure AD Easy Auth · AWS ALB+Cognito · dev auto-auth

STORAGE_MODE=gcs | s3 | azure | local — same Go API, no code changes

Container Apps · PostgreSQL Flex · Azure Blob · ACR · Easy Auth · Azure VM (DIMSE)

GitHub Actions: Go (137+) · Python (617+) · TS · Docker (13) · Cloud Build auto-deploy

aegisimaging.ai · www · api · admin — SSL cert v3